Importance Of QUALITY ENGINEERING In Consumer Electronics

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In a modern world, it is a big challenge for many industries in information technology or manufacturing to maintain quality without compromising on the costs of quality and non-quality. These two together make the total cost towards maintaining quality of a product, and are involved throughout the project's life cycle. The primary challenge here is to determine how to trim the overall costs of quality and non-quality.

There are many definitions of quality set by various gurus around the world, some of which are given below:

"Quality should be aimed at the needs of customers, present and future."

—Dr Edward Deming

"Quality is the degree of excellence at an acceptable price and control of variability at an acceptable cost."

—Robert A. Broh

"Quality is the loss (from function variation and harmful effects) a product causes to society after being shipped, other than any losses caused by its intrinsic functions."

—Dr Genichi Taguchi

Consumer electronic products have huge competition in today's market. Due to this, there is a thin cost margin with good character that draws consumers to buy the merchandise, along with quick and quality after-sales service.

Examples from open source that speak about the impact of cost of non-quality

- German car maker, Volkswagen, recalled all 5561 e-golf battery-electric cars sold in the USA between May 21, 2014 and March 1, 2016 because of faulty battery software that could cause the cars to stall and crash, according to the company's safety recall report filed with National Highway Traffic Safety Administration.
- Customers returning electronics products had cost the USA consumer electronics retailers and manufacturers nearly US$ 17 billion in 2016, an increase of 21 per cent since 2007, according to Accenture research report. These costs include receiving, assessing, repairing, reboxing, restocking and reselling returned products.

There are many factors that may affect costs of quality and non-quality. A few of these are mentioned below.

Cost of quality is the cost of conformance incurred from costs of prevention and appraisal. These include:

- Cost of contract
- Cost of employee training and education
- Cost of auditing
- Cost of maintenance
- Cost of resolving an impediment
- Cost of maintaining company assets like test equipment
- Cost due to field trials
- Cost of employee wages
- Cost of prototype and reviews
- Cost of continuous improvement
- Cost to make infrastructure for logistics and packing (focus on packing technologies)

Cost of non-quality is the cost of non-conformance incurred due to internal and external failures. These include:

- Cost of rework
- Cost of frequent design change
- Cost of requirement change
- Cost of material waste
- Cost of delay in release due to internal failure or regressions
- Cost due to non-conformance in manufacturing process
- Cost due to non-conformance in raw materials like electronic and electrical components
- Cost due to non-conformance in assembly of components
- Cost due to warranty claims
- Cost incurred from customer complaints
- Cost due to return of product
- Cost due to service
- Cost due to lack of service (losing customers)
- Cost due to legal and government rules, when certain regulations are not fol-